

TECHNOLOGY



Designing Microsoft Azure Infrastructure Solutions: AZ:305

Design a Compute Solution



Learning Objectives

By the end of this lesson, you will be able to:

- Evaluate appropriate compute provisioning strategies to optimize resource allocation
- Demonstrate proficiency in selecting and configuring Azure App Service solutions, aligning organizational requirements with platform capabilities
- Analyze and compare various compute technologies available in Azure



Learning Objectives

By the end of this lesson, you will be able to:

- Evaluate containerization options in Azure, including Docker and Azure Kubernetes Service (AKS), to propose an optimized solution for deploying, managing, and scaling containerized applications
- Design and implement serverless solutions using Azure Functions and Logic Apps for demonstrating the ability to automate workflows, integrate services, and address business logic requirements in a scalable and cost-effective manner



Recommend a Solution for Compute Provisioning

Azure Compute Service

Compute refers to the hosting model for the computing resources that the application runs on.



The application may have multiple workloads, and each workload must be evaluated separately.

As a result, a complete solution may incorporate two or more compute services.

Types of Compute Strategies

There are two types of compute strategies:



Lift and shift

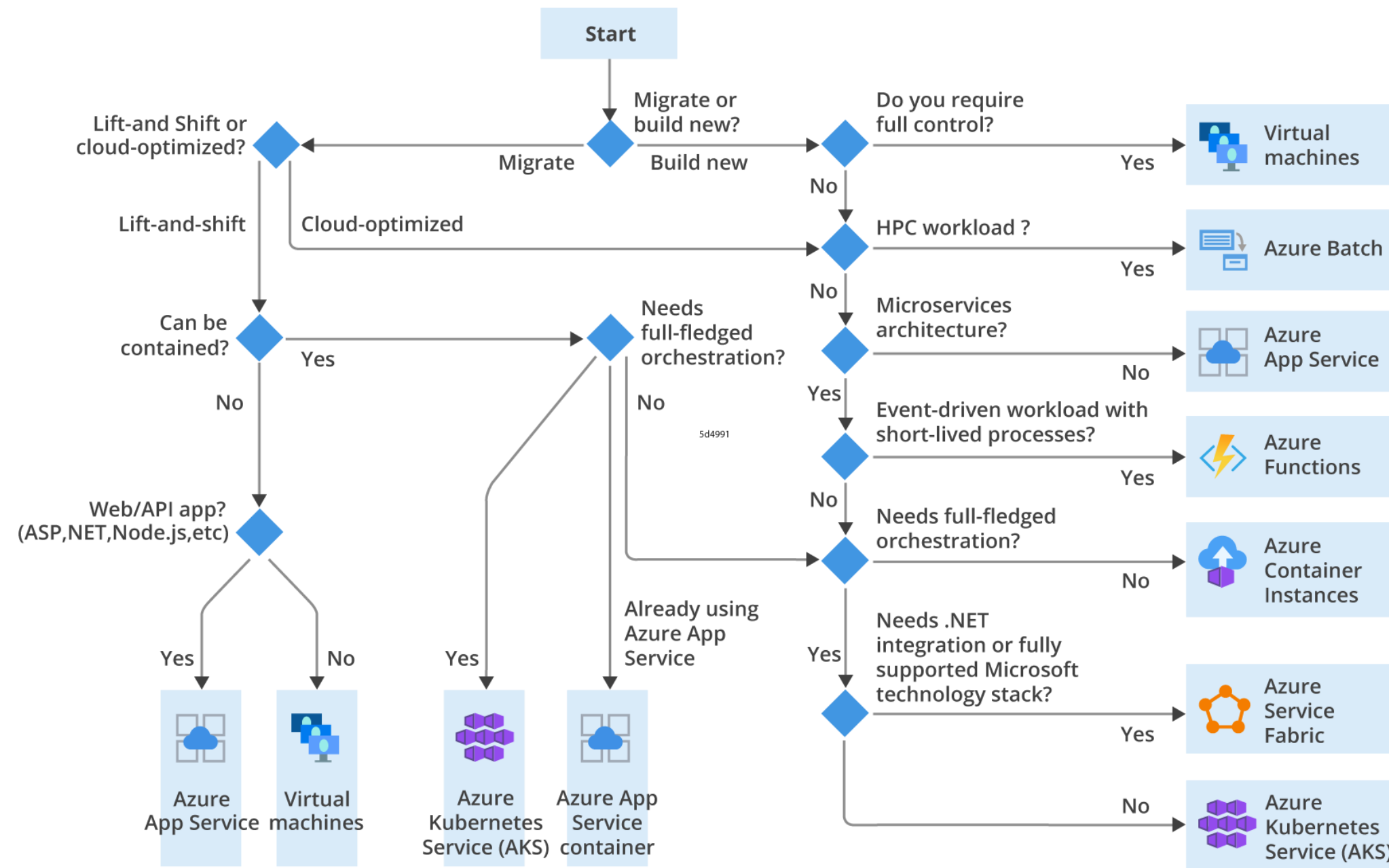
Migrating a workload without redesigning the application or making code changes

Cloud optimized

Taking advantage of cloud-native features and capabilities

Choose a Candidate Service

Use the following flowchart to choose a candidate compute service:



Hosting Models

There are two most common hosting models:

IaaS (Infrastructure as a service)

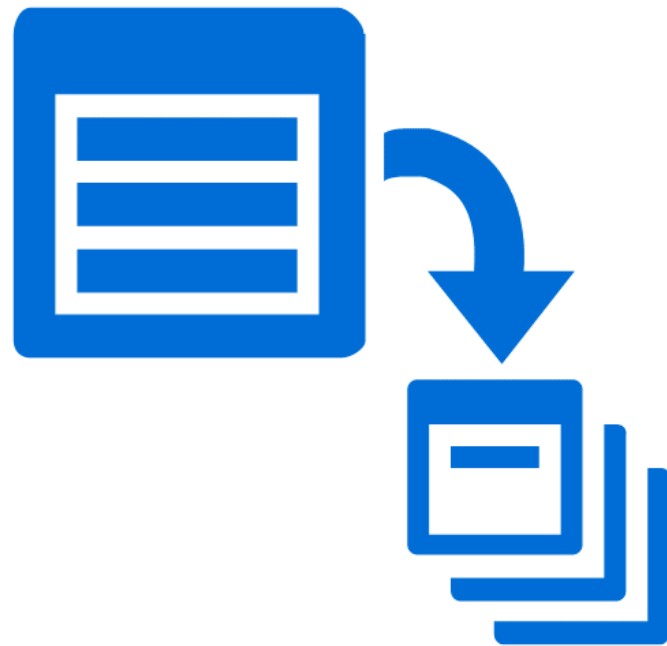
- It lets users provision individual VMs.
- It provides control, flexibility, and portability.

PaaS (Platform as a service)

- It is a complete development and deployment environment in the cloud with resources.
- It enables the delivery of everything from simple cloud-based apps to sophisticated, cloud-enabled enterprise applications.

Batch

It is an Azure compute management platform that lets users run large-scale parallel batch applications in cloud.



End-users may provision a variety of scalable, high-performance resources with ease and at a low cost with the help of Azure Batch.

Features of Batch



It allows users to:

- Conduct large-scale parallel workloads at a reasonable cost by utilizing low-priority virtual machines
- Configure the nodes themselves, given that it supports Docker configuration
- Create pools and run jobs through an easy-to-use code interface with R

Applications of Batch

Load testing software application

Software testers can use Azure Batch service to arrange the provisioning of thousands of virtual machine nodes to test their apps for peak load and consumption.



Applications of Batch

Execute high-compute apps and services

The Azure Batch function scheduling service automates the addition of compute clusters for applications and services that require a lot of processing power.



Applications of Batch

Provide on-demand processing

Organizations can use Azure Batch service to provide on-demand and high-end processing capabilities and infrastructure for apps and services.



Candidate Services

Following are the different candidate services:

App Service

It is a managed service for hosting web apps, mobile app back ends, RESTful APIs, or automated business processes.

Functions

It is a managed PaaS service.

Batch

It is a managed service for running large-scale parallel and high-performance computing (HPC) applications.

Azure Kubernetes
Service

It is a managed Kubernetes service for running containerized applications.

Candidate Services

Following are the different candidate services:

Container Instances

It provides the fastest and simplest way to run a container in Azure without provisioning any virtual machines or adopting a higher-level service.

Virtual machines

It allows deployment and management of VMs inside an Azure virtual network.

Azure Service Fabric

It allows packaging, deployment, and management of scalable and reliable microservices and containers.

Service Limits and Cost

Every service has service limits applicable.

Criteria	Virtual Machines	App Service	Service Fabric	Azure Functions	Azure Kubernetes Service	Container Instances	Azure Batch
Application composition	Agnostic	Applications, containers	Services, guest executables, containers	Functions	Containers	Containers	Scheduled jobs
Density	Agnostic	Multiple apps per instance via app service plans	Multiple services per VM	Serverless	Multiple containers per node	No dedicated instances	Multiple apps per VM
Minimum number of nodes	1	1	5	Serverless	3	No dedicated nodes	1
State management	Stateless or Stateful	Stateless	Stateless or stateful	Stateless	Stateless or Stateful	Stateless	Stateless
Web hosting	Agnostic	Built in	Agnostic	Not applicable	Agnostic	Agnostic	No
Can be deployed to dedicated VNet?	Supported	Supported	Supported	Supported	Supported	Supported	Supported
Hybrid connectivity	Supported	Supported	Supported	Supported	Supported	Not supported	Supported

It is important to consider the cost, the Service Level Agreements (SLA), and the regional availability of the service.

Scalability of Compute Services

Guidance on service to be used for scaling different resources:

Criteria	Virtual Machines	App Service	Service Fabric	Azure Functions	Azure Kubernetes Service	Container Instances	Azure Batch
Autoscaling	Virtual machine scale sets	Built-in service	Virtual machine scale sets	Built-in service	Pod auto-scaling, cluster auto-scaling	Not supported	N/A
Load balancer	Azure Load Balancer	Integrated	Azure Load Balancer	Integrated	Azure Load Balancer or Application Gateway	No built-in support	Azure Load Balancer
Scale limit	Platform image: 1000 nodes per scale set; Custom image: 600 nodes per scale set	30 instances, 100 with App Service Environment	100 nodes per scale set	200 instances per function app	100 nodes per cluster (default limit)	20 container groups per subscription (default limit)	20 core limit (default limit)

Availability of Compute Services

Guidance on service to be used for availing different resources:

Criteria	Virtual Machines	App Service	Service Fabric	Azure Functions	Azure Kubernetes Service	Container Instances	Azure Batch
Multi-region failover	Traffic manager	Traffic manager	Traffic manager and Multi-Region Cluster	Azure Front Door	Traffic manager	Not supported	Not supported

Azure VM vs. Azure VMSS

When to Use Azure VM?

It is used when the user wants to:

Run workloads with varying resource requirements

Host custom applications with specific software requirements

Migrate or hosting legacy applications

Create development and testing environments

Run Windows or Linux-based applications

When to Use Azure VM?

It is used when the user wants to:

Perform
complex
calculations or
simulations

Specify network
configurations
or deploy
custom network
appliances

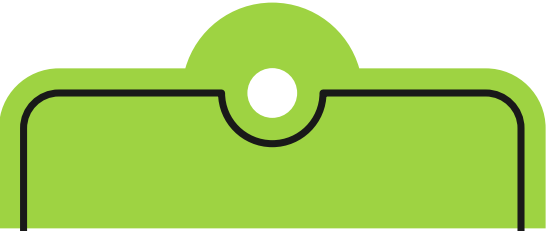
Host databases
or server
applications

Have global
availability

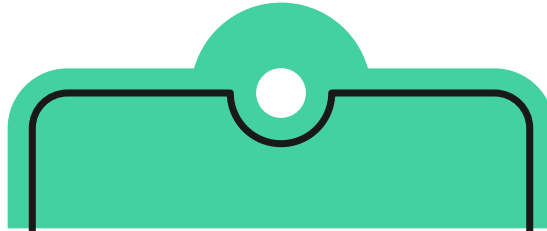
Integrate on-
premises
infrastructure
with the cloud

When to Use Azure VM Scale Sets?

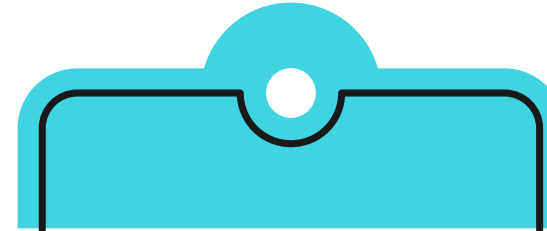
It is used when the user wants to:



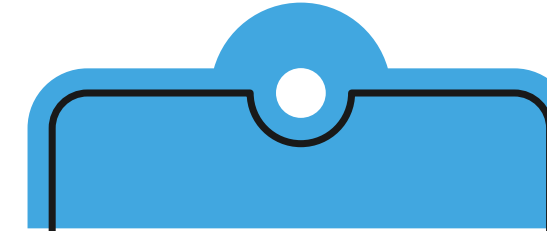
Run workloads with varying resource needs that require automatic scaling



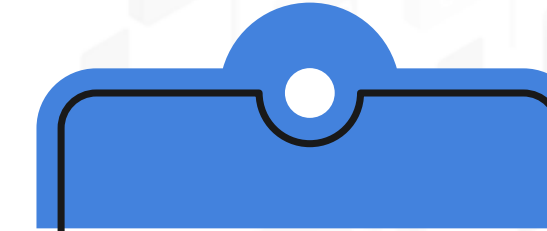
Scale the number of VM instances in or out



Prioritize fault tolerance and high availability



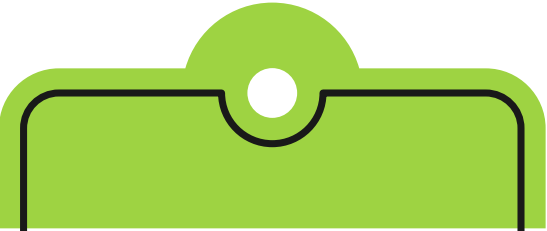
Have a uniform configuration for all VM instances



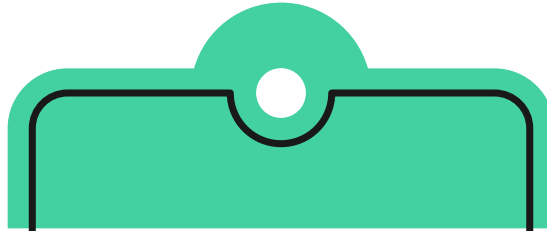
Deploy stateless applications that can scale horizontally

When to Use Azure VM Scale Sets?

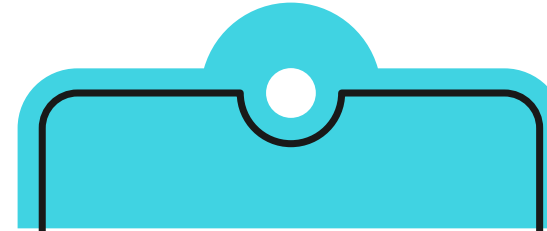
It is used when the user wants to:



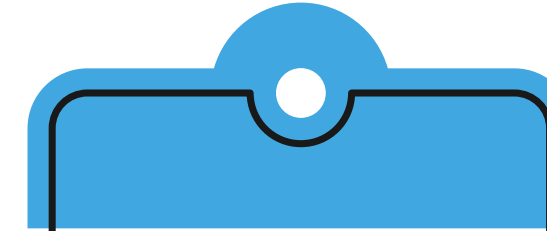
Have efficient load balancing across multiple VM instances



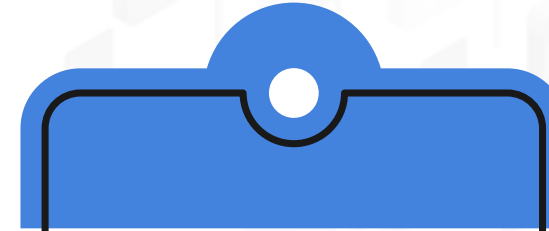
Update applications with minimal downtime



Simplify management of a group of VM instances



Plan for future growth and scalability



Have cost efficiency through optimized resource utilization

Azure VM vs. Azure VMSS

Scenario	Group of virtual machines	Virtual machine scale sets
To add VM instances for changing workload	Manual process to create, configure, and ensure compliance	Automatically create from central configuration
To balance and distribute workloads	Manual process to create and configure Azure load balancer or application gateway	Can automatically create and integrate with Azure load balancer or application gateway
To have high availability and redundancy	Manually create an availability set or distribute and track VMs across availability zones	Automatic distribution of VM instances across availability zones (when zones are selected) and Availability Set
To monitor and then scale virtual machines	Manual monitoring and Azure automation	Autoscale based on host metrics, in-guest metrics, application insights, or schedule

Creating a Virtual Machine



Duration: 10 Min.

Problem Statement:

As an Azure architect, you have been tasked with recommending an Azure computing solution that provides an isolated environment for operating the OS and apps without relying on the underlying host system. Your company will use this solution to deploy an application for testing and production.

Outcome:

This setup ensures that the RDS instance is created with the necessary security and configuration parameters, making it ready for use in your application.

Note: Refer to the demo document for detailed steps: 01_Creating_a_Virtual_Machine

ASSISTED PRACTICE

Assisted Practice: Guidelines

Steps to be followed:

1. Create a virtual machine



Manage VM Sizes



Duration: 10 Min.

Problem Statement:

As an Azure Architect, you have been tasked with demonstrating the various VM sizes available in Azure for running your apps and workloads.

Outcome:

This setup ensures that the RDS instance is created with the necessary security and configuration parameters, making it ready for use in your application.

Note: Refer to the demo document for detailed steps: 02_Manage_VM_Sizes

UNASSISTED PRACTICE

Unassisted Practice: Guidelines

Steps to be followed:

1. Manage VM size



Recommend a Solution for App service

Azure App Service

It's a unique cloud service enabling users to quickly develop enterprise-ready web and mobile apps for any platform, deploying them on a scalable and reliable cloud infrastructure.



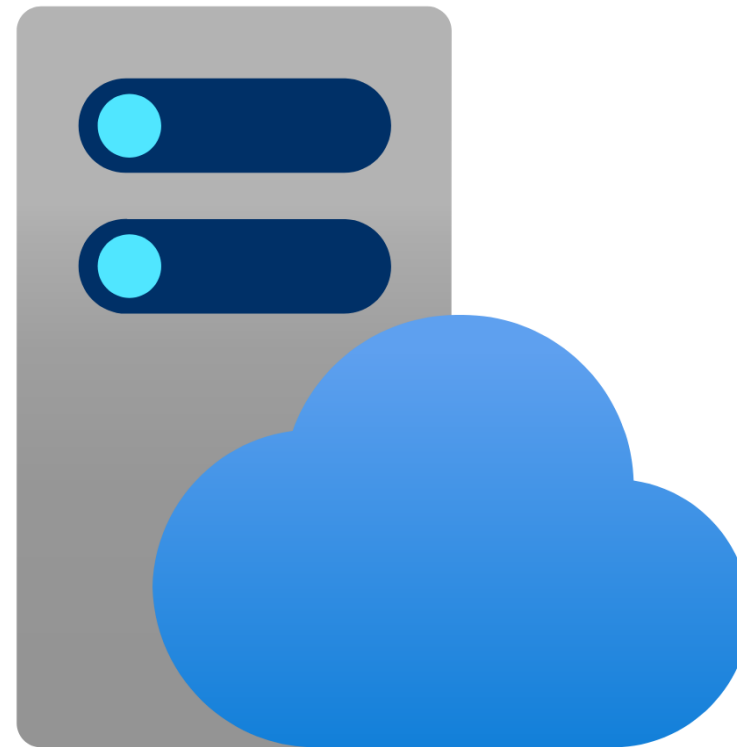
Azure App Service: High Availability

The Azure App Service is a PaaS offering from Microsoft that enables a user to build, deploy, and scale web applications.



Azure App Service Plan

It defines a set of compute resources for a web app to run.



It determines the capabilities and resources available to its web apps.



Azure App Service Plan

Web apps can share the same App Service plan.



Isolate the app into a new App Service plan when:

- The app is resource-intensive.
- A user wants to scale the app independently from other apps.
- The app needs resources in a different geographical region.

Azure App Service Plans Scalability

It refers to the adaptability of the system to the modified amount of workload or traffic.



It allows increasing or decreasing the resources for a user's application.

Scale In or Out

The autoscale feature allows the user to scale in and out the number of instances of their app service plan based on the schedules they set or the metrics they define.

In the Basic Tier, users can scale up to 3 instances.

In Isolated (App Service Environment), there can be 100 instances.

In the Premium Tier, the maximum is 20 instances.

In the Standard Tier, users can scale up to 10 instances.

Scale Up or Down

The following are the different types of tiers for scaling up or down:

Basic



Standard



Premium

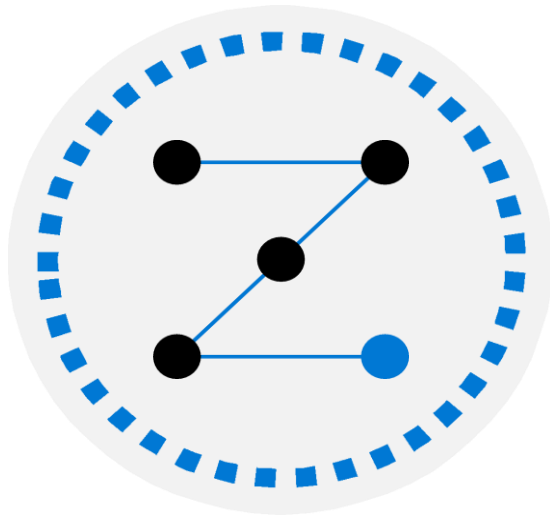


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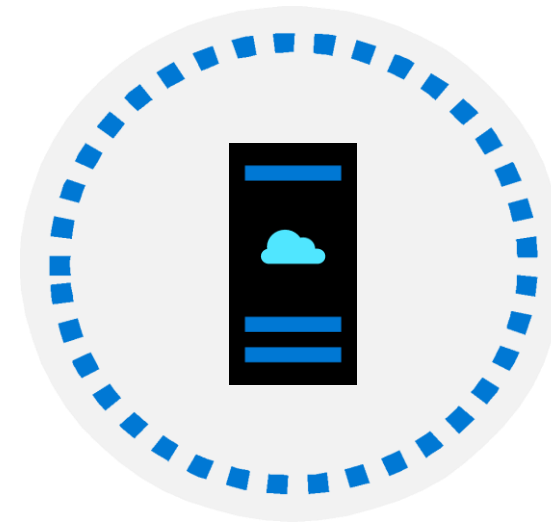
Determine Appropriate Compute Technologies

Choosing an Azure Compute Option for Microservices

Below are the popular approaches to a Microservices architecture:



A service orchestrator that manages services running on dedicated nodes (VMs)



A serverless architecture using functions as a service (FaaS)



Service Orchestrators

An orchestrator handles tasks related to deploying and managing a set of services.



Tasks include:

- Placing services on nodes
- Monitoring the health of services
- Restarting unhealthy services
- Load balancing network traffic across service instances
- Service discovery
- Scaling the number of instances of a service
- Applying configuration updates

Containers with Orchestration

Consider the orchestration services on the Azure Platform:

Azure Kubernetes Service (AKS):

It provisions Kubernetes and exposes the Kubernetes API endpoints.

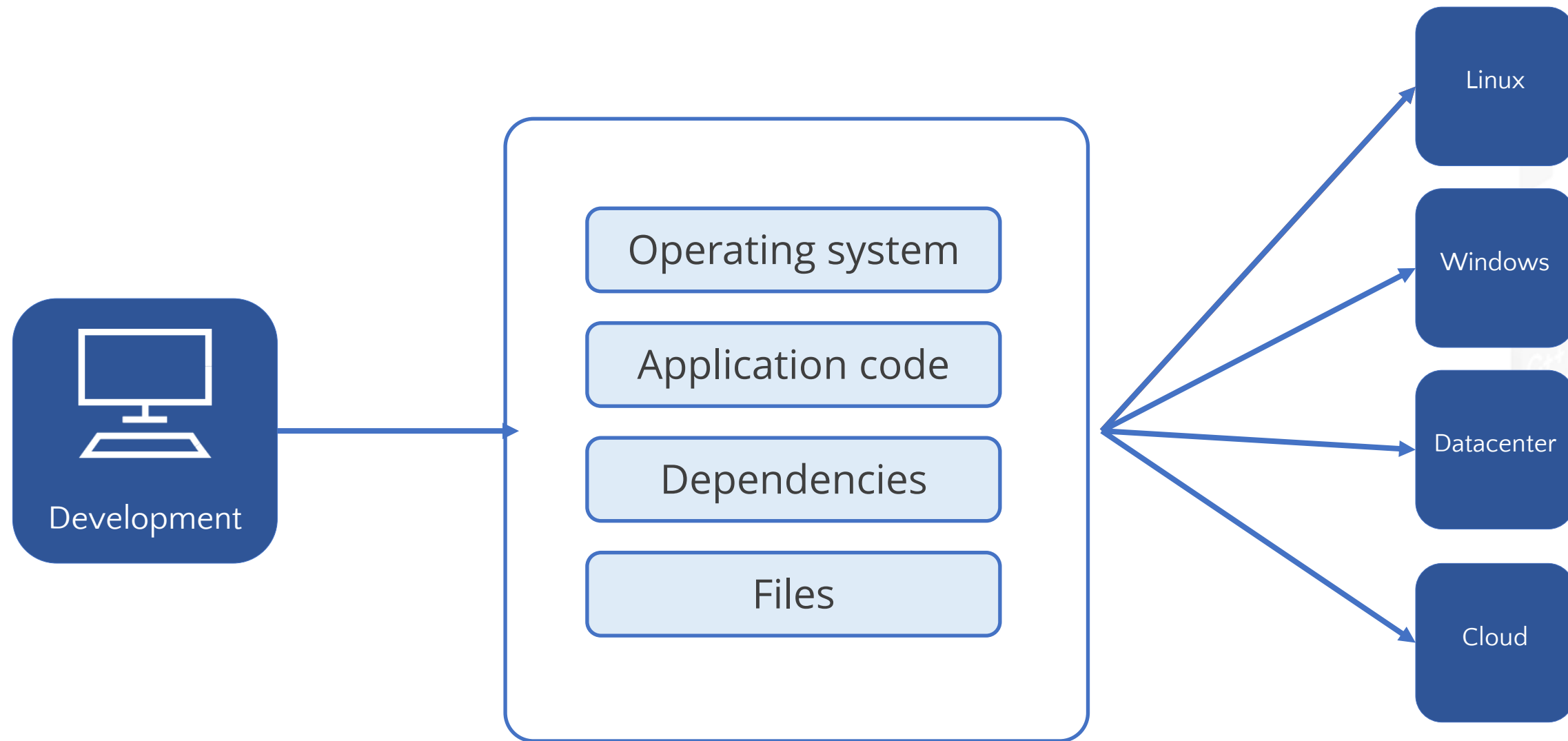
Service Fabric:

It is a distributed systems platform for packaging, deploying, and managing microservices.

Note: Docker Enterprise Edition and Mesosphere DC/OS can run in an IaaS environment on Azure.

Containers

It is a loosely isolated environment that allows you to build and run software packages.



Containers

Containers offer benefits that are specifically relevant to microservices:

Portability

A container image is a standalone package that runs without installing libraries or other dependencies.

Agility

Containers can start and stop quickly. They spin up new instances to handle more load or to recover from node failures.

Density

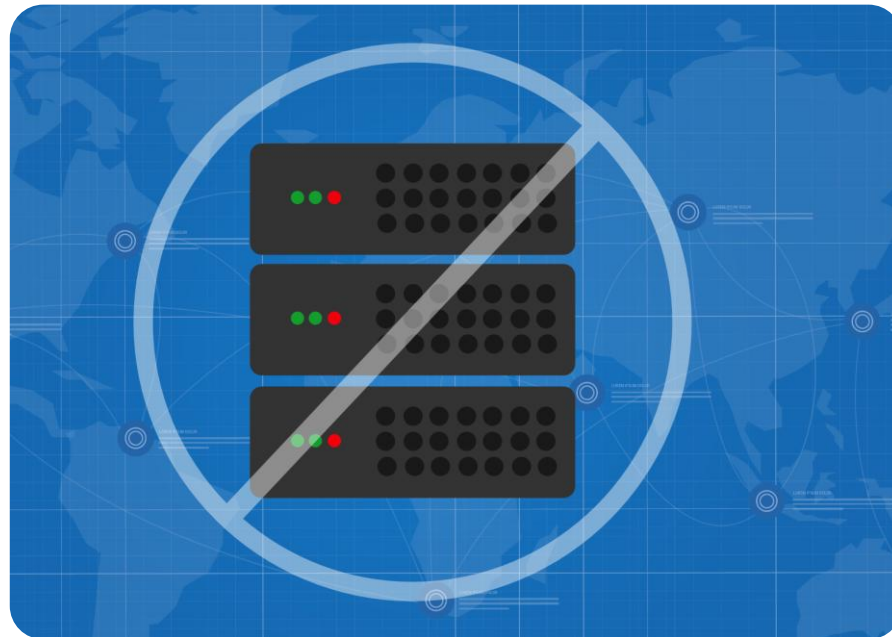
Packing multiple containers onto a single node is useful when the application consists of many small services.

Resource isolation

The amount of memory and CPU that is available must be limited to a container to ensure that a runaway process does not exhaust the host resources.

Serverless

With a serverless architecture:



- Users do not manage the VMs or the virtual network infrastructure.
- They deploy code, and the hosting service handles the putting and executing of that code onto a VM.

This approach suits the small, granular functions that are coordinated using event-based triggers.

Serverless

Azure Functions is a serverless compute service that supports various function triggers, including:

HTTP requests

Service Bus
queues

Event Hubs
events



Example: A message being placed in a queue might trigger a function that reads from the queue and processes the message.

Orchestrator vs. Serverless

These are the difference between orchestrator and serverless:

Manageability

Flexibility and control

Portability

Application integration

Cost

Scalability

- Serverless applications are easy to manage as the platform manages all the compute resources.
- However, with an orchestrator, a user must keep an eye on load balancing, CPU, memory usage, and networking.

Orchestrator vs. Serverless

These are the difference between orchestrator and serverless:

Manageability

Flexibility and control

Portability

Application integration

Cost

Scalability

- In a serverless architecture, users give up some degree of control because management details are abstracted.
- An orchestrator gives the user good control over configuring and managing user services and the cluster.

Orchestrator vs. Serverless

These are the difference between orchestrator and serverless:

Manageability

Flexibility and control

Portability

Application integration

Cost

Scalability

- Orchestrators can run on-premises or in multiple public clouds.

Orchestrator vs. Serverless

These are the difference between orchestrator and serverless:

Manageability

Flexibility and control

Portability

Application integration

Cost

Scalability

- Building a complex application using serverless is difficult. Users can opt for Azure Logic Apps to coordinate a set of Azure Functions.

Orchestrator vs. Serverless

These are the difference between orchestrator and serverless:

Manageability

Flexibility and control

Portability

Application integration

Cost

Scalability

- With a serverless application, the user pays for the actual compute resources consumed.
- With an orchestrator, the user pays for the VMs running in the cluster.

Orchestrator vs. Serverless

These are the difference between orchestrator and serverless:

Manageability

Flexibility and control

Portability

Application integration

Cost

Scalability

- An orchestrator scales out by increasing the number of service instances running in the cluster, or, adding additional VMs to the cluster.
- Serverless compute platforms like Azure Functions scale automatically.

Recommend a Solution for Containers

Azure Container Instance

It provides the quickest and easiest method to run a container in Azure, eliminating the need to manage virtual machines.



- It does not require IaaS provisioning.
- It does not require the adoption of a higher-level service.
- It supports Linux and Windows containers.
- It supports direct mounting of Azure File Shares.

Azure Container Instance

Azure Container Instance has the following features:



- Run Containers without managing servers
- Increase agility with containers on demand
- Secure applications with hypervisor isolation

A container can be provisioned with a public IP address and DNS name.

When to Use Azure Container Instances

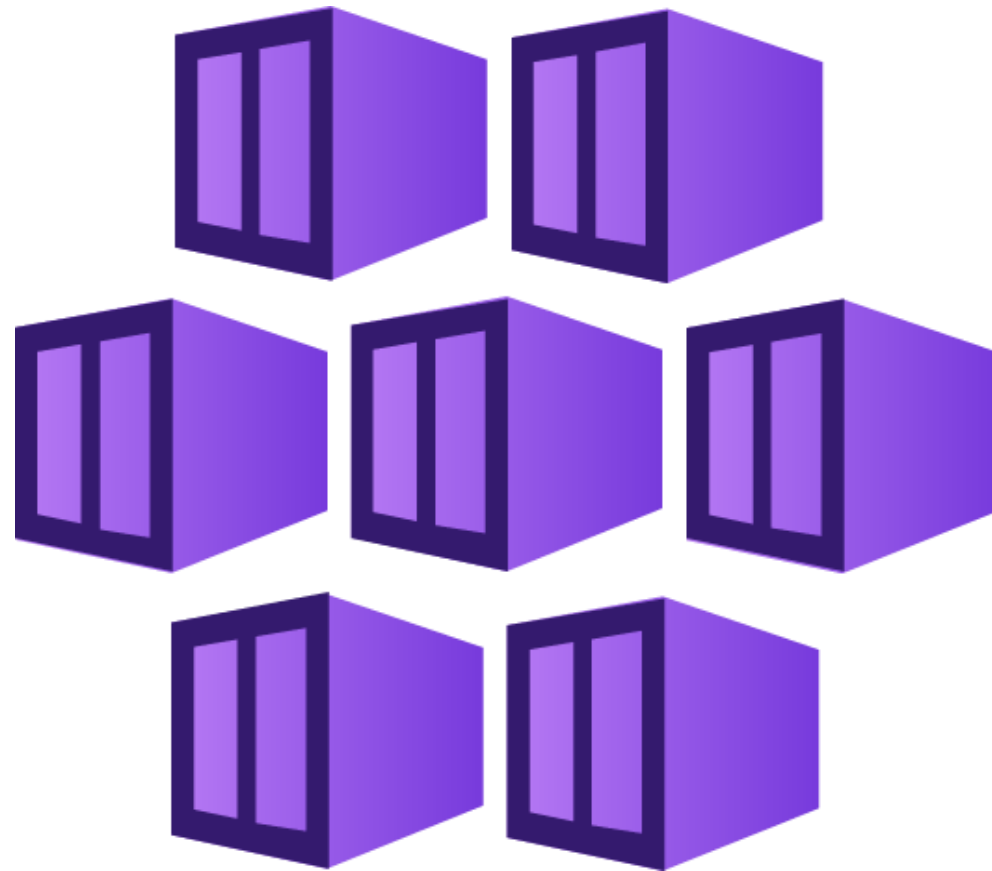
Feature	Description
Fast startup times	Containers can start in seconds without the need to provision and manage VMs.
Public IP connectivity and DNS name	Containers can be directly exposed to the internet with an IP address and a fully qualified domain name (FQDN).
Hypervisor-level security	Container applications are as isolated in a container as they would be in a VM.
Custom sizes	Container nodes can be scaled dynamically to match the actual resource demands of an application.

When to Use Azure Container Instances

Feature	Description
Persistent storage	Containers support direct mounting of Azure Files shares.
Linux and Windows containers	The same API is used to schedule both Linux and Windows containers.
Co-scheduled groups	Container Instances support the scheduling of multi-container groups that share host machine resources.
Virtual network deployment	Container Instances can be deployed into an Azure virtual network.

Azure Kubernetes Service

Azure Kubernetes Service (AKS) manages your hosted Kubernetes environment and makes it simple to deploy and manage containerized applications in Azure.



Azure Kubernetes Service

The Azure Kubernetes Service provides the following features:

Manages container-based applications

- It handles networking and storage needs concurrently.
- It focuses on application workloads instead of infrastructure components.

Describes the applications declaratively

- YAML files can be used to describe the application.
- Kubernetes handles management and deployment.

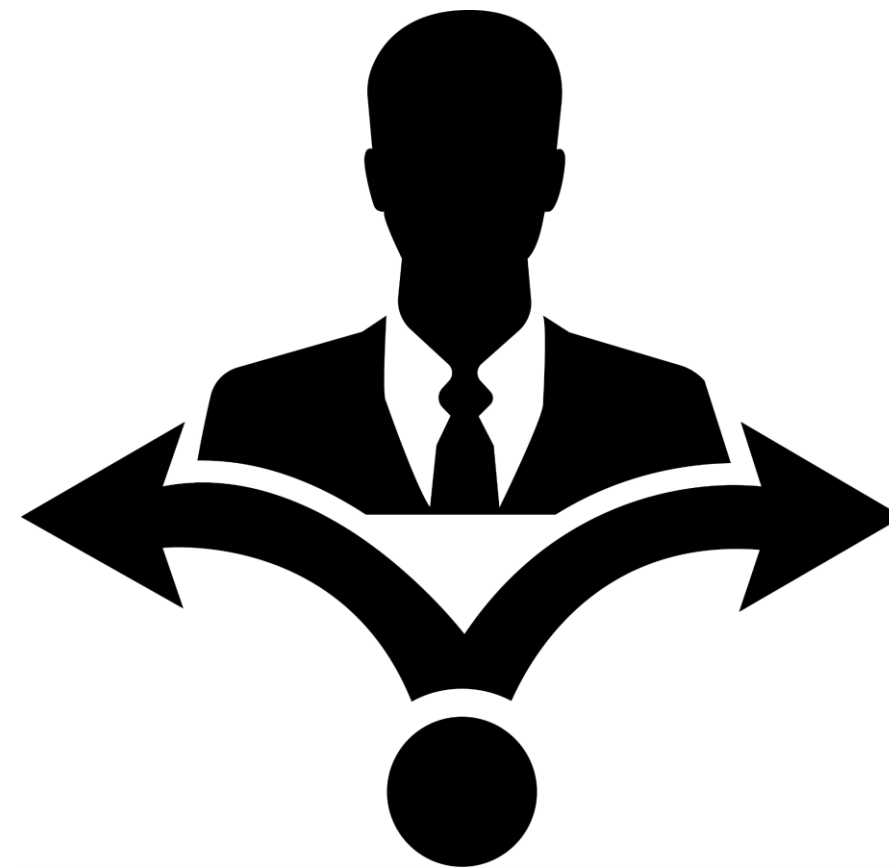
Makes it easier to orchestrate large solutions using a variety of containers

- Application containers
- Storage containers
- Middleware containers

When to Use Azure Kubernetes Service

Users would approach the decision based on:

Greenfield deployment,
which allows user to
evaluate AKS based on
default features



Lift and Shift Containers,
which force users to
consider features that
best support their
migration

When to Use Azure Kubernetes Service

These are various features considered while deciding:

Feature	Considerations and Decisions
Identity and security	AD integration
Logging and monitoring	Azure Monitor for containers and custom monitoring solutions
Auto scaling	Manual and auto, scheduled and programmatically, horizontal and vertical
Cluster node upgrade	Azure manages cluster upgrade and software updates
GPU support	GPU-enabled node pools

When to Use Azure Kubernetes Service

These are various features considered while deciding:

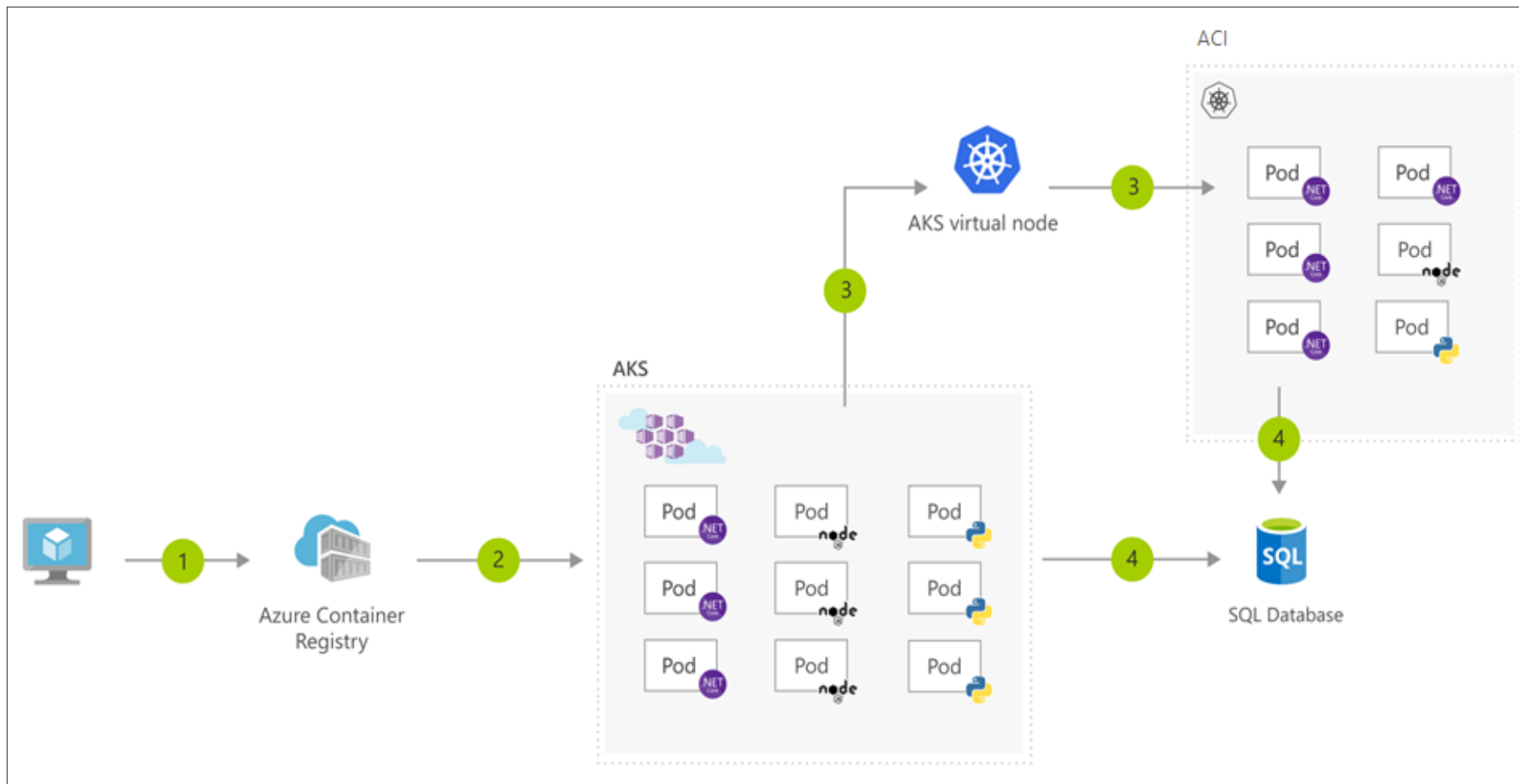
Feature	Considerations and Decisions
Storage volumes	Static and dynamic storage volumes
Virtual Network (VNet)	Deployed into existing VNet
HTTP routing	HTTP application routing support is an add on
Docker image	Default support for docker file image support
Private Container Registry	ACR (Azure Container Registry) Public and Private repositories

Recommendation

For scenarios where users need full container orchestration, including service discovery across multiple containers, automatic scaling, and coordinated application upgrades, use Azure Kubernetes Service (AKS).



Bursting from AKS with ACI



- A user registers a container in the Azure Container Registry.
- AKS virtual node, a Virtual Kubelet implementation, provisions pods inside ACI from AKS when traffic spikes.
- AKS and ACI containers write to the shared datastore.

Creating a Container Instance



Duration: 10 Min.

Problem Statement:

As an Azure architect, you have been asked to provide your organization with an Azure solution that can provide the quickest and most straightforward approach to host a container in Azure, without the need to manage any virtual machines or adopt a higher-level service.

Outcome:

This setup ensures that the RDS instance is created with the necessary security and configuration parameters, making it ready for use in your application.

Note: Refer to the demo document for detailed steps: [03_Creating_a_Container_Instance](#)

ASSISTED PRACTICE

Assisted Practice: Guidelines

Steps to be followed:

1. Create a container instance



Creating a Container Registry



Duration: 10 Min.

Problem Statement:

Demonstrate ACR, a private registry for Docker container images hosted on Azure.

Outcome:

This setup ensures that the RDS instance is created with the necessary security and configuration parameters, making it ready for use in your application.

Note: Refer to the demo document for detailed steps: [04_Creating_a_Container_Registry](#)

ASSISTED PRACTICE

Assisted Practice: Guidelines

Steps to be followed:

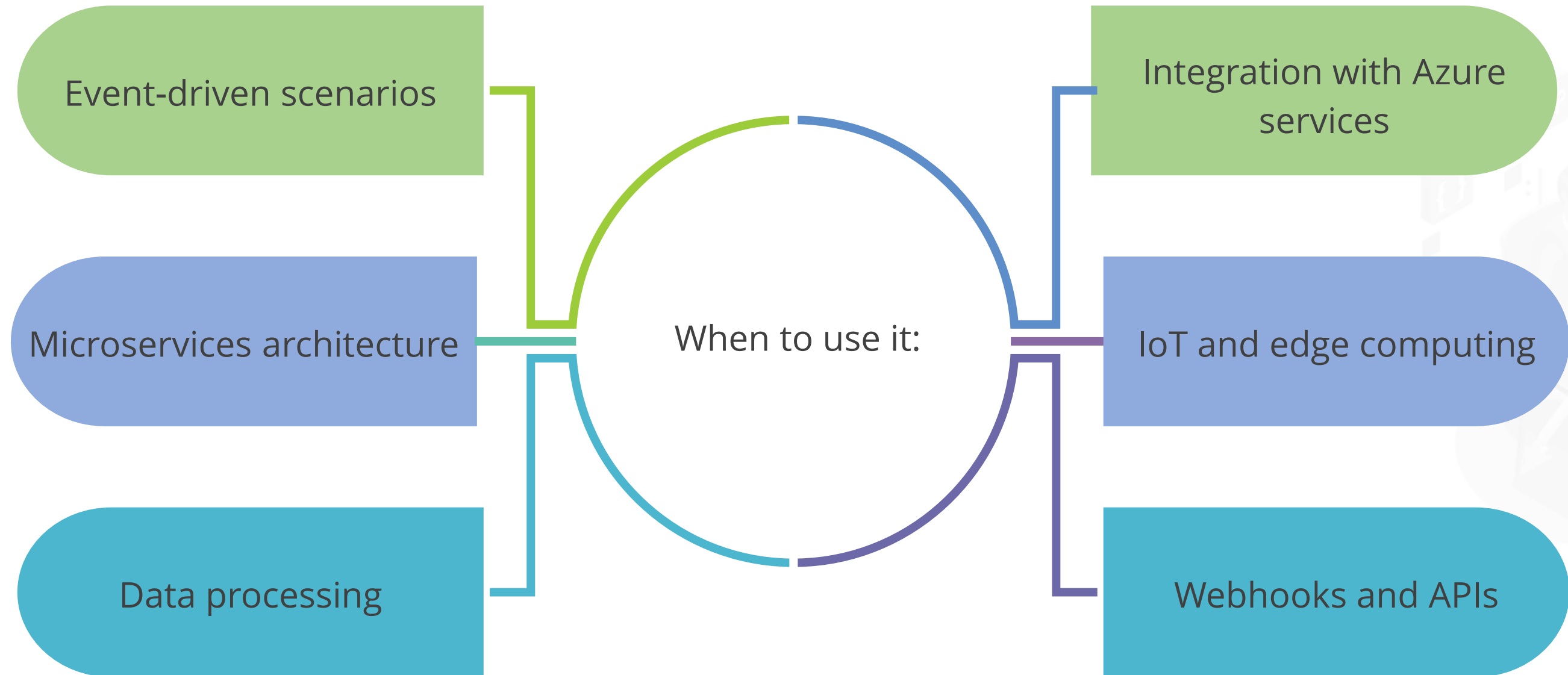
1. Create a container registry



Design for Azure Functions and Azure Logic Apps

Azure Functions

It is a serverless solution for every application.



Azure Functions

Here are the factors to consider for Azure functions:

Authentication
and authorization

Billing and
consumption
model

Stateless
execution

Dependency
management

Logging and
monitoring

Scaling

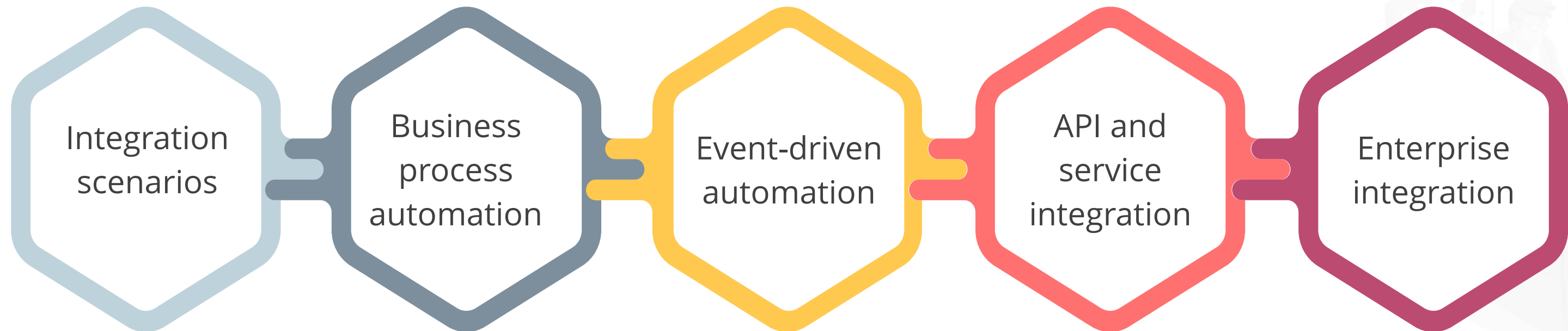
Execution timeout

Cold start latency

Azure Logic App

It is a cloud-based service that allows users to create, automate, and orchestrate workflows to integrate systems, services, and applications.

Following are the circumstances to use logic app:



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The following workflow explains it:

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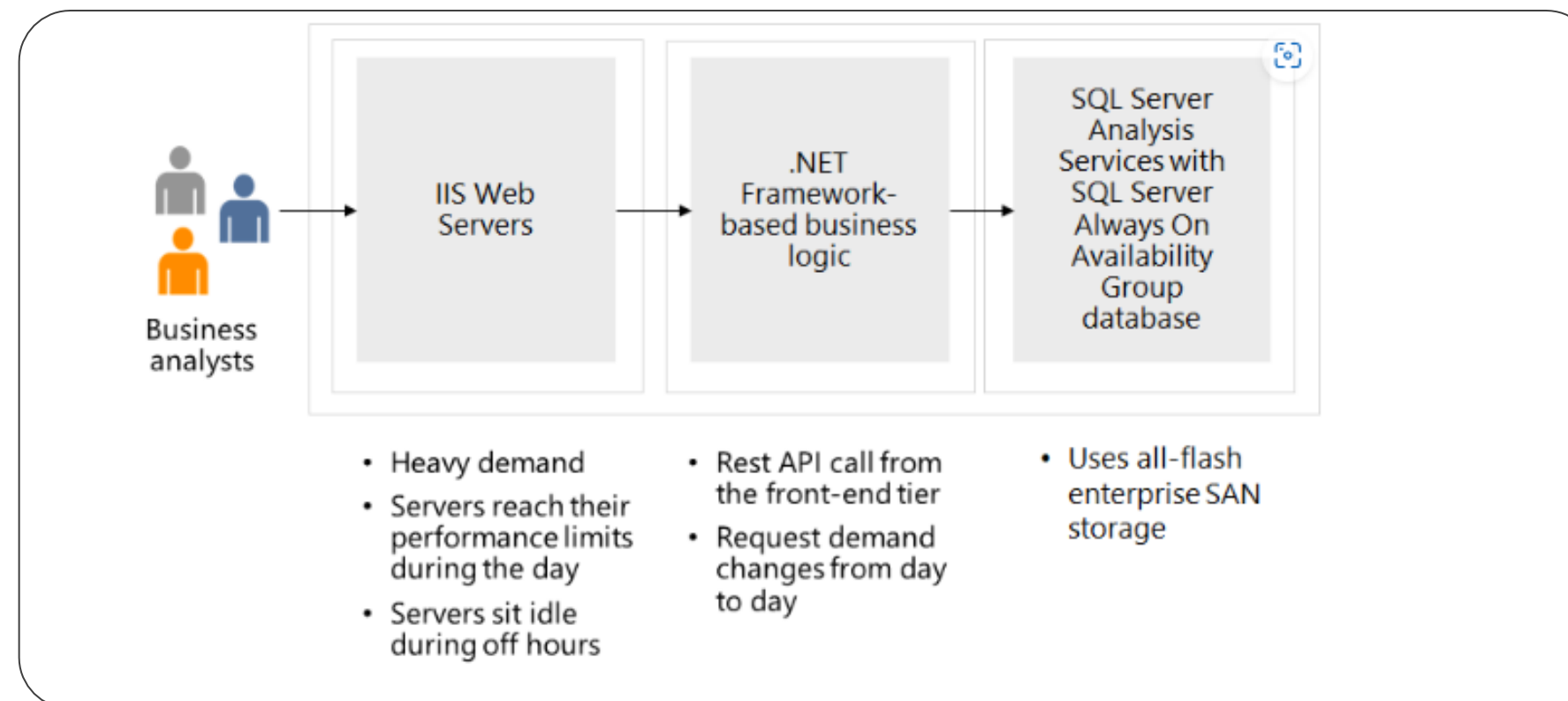


Case Study

Design for Compute Solutions

Scenario

“ABC” Company would like to migrate its product catalog application to the cloud. This application has a traditional 3-tier configuration using SQL Server as the data store. The IT team hopes you can help modernize the application. They have provided this diagram and several areas that could be improved.



Design for Compute Solutions

Scenario

- The frontend application is a .NET core-based web app. During peak periods, 1750 customers visit the website each hour.
- The application runs on Internet Information Service (IIS) web servers in a front-end tier. This tier handles all customer requests for purchasing products. During the latest holiday sale, the front-end servers reached their performance limits, and page loads were lengthy. The IT team has considered adding more servers, but the servers are often idle during off-hours.

Design for Compute Solutions

Scenario

- The middle tier hosts the business logic that processes customer requests. These requests are often for help desk support. Support requests are queued, and lately, the wait times have been very long. Customers are offered email rather than waiting for a representative. However, many customers seem frustrated and are disconnecting rather than waiting. Customer requests are 75-125 per hour.
- The back-end tier uses an SQL Server database to store customer orders. Currently, the back-end database servers are performing well.
- While high availability is a concern due to legal requirements, the company must keep all the resources in a single region.

Design for Compute Solutions

Requirements

- **Front-end tier:** Which Azure compute service would you recommend for the front-end tier? Explain why you decided on your solution.
- **Middle tier:** Which Azure compute service would you recommend for the middle tier? Explain why you decided on your solution.

How are you incorporating the well-architected framework pillars to produce a high-quality, stable, and efficient cloud architecture?

Key Takeaways

- AKS is a managed service for running large-scale parallel and high-performance computing (HPC) applications.
- Azure Functions is a serverless compute service that supports various function triggers.
- A container can be provisioned with a public IP address and DNS name.
- The DSC extension handler grabs the configuration and implements the state on the target VM.





Knowledge Check

What is the simplest way to run an Azure container?

- A. Container Instance, without requiring an IaaS provisioning
- B. Container Instance, without having to adopt a higher-level service
- C. Both option A and B
- D. None of the above



Knowledge Check

01

What is the simplest way to run an Azure container?

- A. Container Instance, without requiring an IaaS provisioning
- B. Container Instance, without having to adopt a higher-level service
- C. Both option A and B
- D. None of the above



The correct answer is **C**

Container Instance is the simplest way to run a container in Azure without requiring an IaaS provisioning, without having to manage any virtual machines, and without having to adopt a higher-level service.

What can a user do with a serverless architecture?

- A. Deploy code
- B. Manages VM
- C. Manages network infrastructure
- D. All of the above



Knowledge Check

02

What can a user do with a serverless architecture?

- A. Deploy code
- B. Manages VM
- C. Manages network infrastructure
- D. All of the above



The correct answer is **A**

With a serverless architecture, a user can deploy code, and the hosting service handles the putting and executing of that code onto a VM.

**Knowledge
Check**

03

Which Azure service is best suited for automating workflows, integrating services, and addressing business logic requirements in a serverless manner?

- A. Azure App Service
- B. Azure Functions
- C. Azure Kubernetes Service (AKS)
- D. Azure Logic Apps



Knowledge
Check

03

Which Azure service is best suited for automating workflows, integrating services, and addressing business logic requirements in a serverless manner?

- A. Azure App Service
- B. Azure Functions
- C. Azure Kubernetes Service (AKS)
- D. Azure Logic Apps



The correct answer is **D**

Azure Logic Apps is designed for serverless workflow automation, providing a visual design interface to seamlessly integrate services and implement business logic without the need for infrastructure management.

Implement Web App

Duration: 25 min.



Project Agenda: To evaluate the use of Azure Web apps for hosting a website

Objective: You have been given a project to evaluate the use of Azure Web apps for hosting Edu-sim's websites, currently hosted in the company's on-premises data centers. The websites are running on Windows servers using the PHP runtime stack. You also need to determine how you can implement DevOps practices by leveraging Azure web app deployment slots.

Implement Web App

Duration: 25 min.

Perform the following:

1. Create an Azure web app
2. Create a staging deployment slot
3. Configure web app deployment settings
4. Deploy code to the staging deployment slot



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Thank You